

Logos Experiment 12

Longer Training for the Best Efficiency Model

This note documents Experiment 12: the wider-shallower, weight-tied, SwiGLU decoder-only Transformer trained for 3000 iterations. The purpose was to test whether the best efficiency-focused architecture from Experiment 9 could close the quality gap with the best quality model after longer training.

Item	Value
Run name	logos_exp12_wider_shallower_weight_tying_swiglu_longer_train
Model family	Decoder-only causal language model
Architecture	Wider-Shallower + Weight Tying + SwiGLU
Dataset	Original OpenWebText subset
Tokenizer	GPT-2 tokenizer
Training objective	Causal next-token prediction
Training budget	3000 iterations
Main question	Can the smaller/faster Exp 9 architecture reach near-Exp 11 quality when trained longer?

1. Research Question

Experiment 9 gave the strongest 1500-iteration efficiency result: only 18.54M parameters, the smallest checkpoint, fast training, and lower energy, while still beating the original baseline in validation quality. Experiment 12 tests whether that same architecture improves substantially when trained for 3000 iterations.

Hypothesis: longer training should reduce validation loss and perplexity while preserving the model size, checkpoint size, VRAM advantage, and efficiency profile of Experiment 9.

2. Model Configuration

Component	Value
Layers	4
Embedding dimension	288
Attention heads	9
Head dimension	32
Context length	256 tokens
Batch size	64
Tokens per step	16,384
Feed-forward layer	SwiGLU
SwiGLU hidden size	768
Normalization	LayerNorm
Weight tying	Active
Trainable parameters	18,542,688

3. Final Results

Metric	Value
Best iteration	3000
Best validation loss	5.3469

Metric	Value
Final validation loss	5.3470
Final validation perplexity	210.00
Checkpoint size	70.76 MB
Peak VRAM	10.54 GB
Training time	228.54 sec = 3.81 min
Training speed near end	about 216k tokens/sec
Generation latency	12.06 ms/token
Average GPU power	258.97 W
Estimated GPU energy	16.29 Wh

4. Comparison with Experiment 9

Experiment 12 keeps the same architecture as Experiment 9 but doubles the training budget from 1500 to 3000 iterations.

Metric	Exp 9: 1500 iters	Exp 12: 3000 iters	Result
Parameters	18.54M	18.54M	Same
Best val loss	5.7388	5.3469	Improved
Final val loss	5.7490	5.3470	Improved
Perplexity	313.87	210.00	Improved
Checkpoint size	70.76 MB	70.76 MB	Same
Peak VRAM	10.54 GB	10.54 GB	Same
Training time	132.14 sec	228.54 sec	Higher
Energy	9.06 Wh	16.29 Wh	Higher

5. Comparison with Experiment 11

Experiment 11 remains the best pure-quality model. Experiment 12 is slightly worse in validation loss and perplexity, but is smaller, faster, lower-VRAM, and lower-energy.

Metric	Exp 11: best quality	Exp 12: best efficiency	Advantage
Parameters	19.24M	18.54M	Exp 12
Final val loss	5.3123	5.3470	Exp 11
Perplexity	202.82	210.00	Exp 11
Checkpoint size	73.45 MB	70.76 MB	Exp 12
Peak VRAM	11.18 GB	10.54 GB	Exp 12
Training time	280.86 sec	228.54 sec	Exp 12
Training energy	19.48 Wh	16.29 Wh	Exp 12

6. Quantified Improvements

Comparison	Result
Final validation loss vs Exp 9	Improved by 0.4020
Perplexity vs Exp 9	Reduced by about 33.10%
Parameters vs original baseline	Reduced by about 42.34%

Comparison	Result
Checkpoint size vs original baseline	Reduced by about 42.34%
Perplexity vs original baseline	Reduced by about 34.05%
Training time vs Exp 11	About 18.63% faster
Training energy vs Exp 11	About 16.38% lower

7. Scientific Interpretation

Experiment 12 confirms that the wider-shallower, weight-tied, SwiGLU architecture was also undertrained at 1500 iterations. Longer training produced a major validation-quality gain while keeping the model size and checkpoint size unchanged.

Although it does not beat Experiment 11 in pure validation quality, it gets close while being smaller, faster, lower-VRAM, and lower-energy. This makes it the best overall efficiency model in the Logos study so far.

8. Conclusion

Experiment 12 is the current best efficiency-focused Logos model. It achieves a final validation loss of 5.3470 and perplexity of 210.00 with only 18.54M parameters and a 70.76 MB checkpoint. It is slightly less accurate than Experiment 11 but offers a stronger speed, memory, storage, and energy tradeoff.

9. Final Status After Experiment 12

Category	Winner
Best quality model	Experiment 11: Weight Tying + SwiGLU, 3000 iterations
Best efficiency model	Experiment 12: Wider-Shallower + Weight Tying + SwiGLU, 3000 iterations
Best checkpoint size	Experiment 9/10/12 family: about 70.76 MB
Best validation loss so far	Experiment 11: 5.3123
Best efficiency-quality compromise	Experiment 12